Chapter 4. Linear Regression

Lecture 6-1 (simple and multiple regression, inference, F-test, qualitative predictors, Gauss–Markov, interactions, polynomials)

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1 Simple Linear Regression

True regression functions are never exactly linear, yet the linear model is very useful. Advertising data: is budget related to sales, how strongly, and which media contribute?

1.1 Model and least squares

$$Y = \beta_0 + \beta_1 X + \varepsilon, \qquad \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

With residuals $e_i = y_i - \hat{y}_i$, RSS $= \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$.

Theorem: least-squares estimates

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Proof [Supplement]. $\partial \mathrm{RSS}/\partial \beta_0 = -2\sum(y_i - \beta_0 - \beta_1 x_i) = 0$ gives $\beta_0 = \bar{y} - \beta_1 \bar{x}$. Substituting, $\partial \mathrm{RSS}/\partial \beta_1 = -2\sum(x_i - \bar{x})\{(y_i - \bar{y}) - \beta_1(x_i - \bar{x})\} = 0$ (using $\sum(x_i - \bar{x}) = 0$) gives $\hat{\beta}_1$. RSS is a convex quadratic, so this is the minimum. ■

Advertising: $\widehat{\texttt{sales}} = 7.03 + 0.0475 \times \texttt{TV}$.

1.2 Accuracy of the coefficient estimates

Lines fitted to repeated samples from $f(X) = 2 + 3X$ scatter around it; the standard error measures this:

Theorem: standard errors

If the errors are uncorrelated with common variance $\sigma^2 = \mathrm{Var}(\varepsilon)$ and the x_i are fixed,

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad \mathrm{SE}(\hat{\beta}_0)^2 = \sigma^2\Big[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\Big].$$

Proof [Supplement]. $\hat{\beta}_1 = \sum_i w_i y_i$ with $w_i = (x_i - \bar{x})/S_{xx}$, $S_{xx} = \sum(x_i - \bar{x})^2$; so $\mathrm{Var}\hat{\beta}_1 = \sigma^2 \sum w_i^2 = \sigma^2/S_{xx}$. Next $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $\mathrm{Cov}(\bar{y}, \hat{\beta}_1) = \frac{\sigma^2}{n}\sum w_i = 0$, so $\mathrm{Var}\hat{\beta}_0 = \sigma^2/n + \bar{x}^2\sigma^2/S_{xx}$. ■

σ is estimated by the **residual standard error** $\mathrm{RSE} = \sqrt{\mathrm{RSS}/(n-2)}$; large S_{xx} gives a precise slope.

1.3 Confidence intervals and tests

A $100(1-\alpha)\%$ confidence interval covers the true parameter with probability $1-\alpha$ under repeated sampling:

$$\hat{\beta}_1 \pm t_{1-\alpha/2,\,n-2} \cdot \mathrm{SE}(\hat{\beta}_1) \quad (\approx \hat{\beta}_1 \pm 2\,\mathrm{SE}(\hat{\beta}_1) \text{ for } 95\%).$$

Advertising: $[0.042, 0.053]$.

$H_0 : \beta_1 = 0$ (no relationship) versus $H_A : \beta_1 \neq 0$:

$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)} \;\sim\; t_{n-2} \text{ under } H_0 \text{ (Gaussian errors)},$$

p-value $= P(|T| \geq |t|)$.

	Coefficient	Std. error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

1.4 Overall accuracy: RSE and R^2

$\text{RSE} = \sqrt{\frac{1}{n-2}\sum(y_i - \hat{y}_i)^2}$ estimates $\text{sd}(\varepsilon)$ (lack of fit, in units of Y);

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}, \qquad \text{TSS} = \sum(y_i - \bar{y})^2,$$

the fraction of variance explained.

Theorem: $R^2 = r^2$ in simple regression

With $r = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2}\sqrt{\sum(y_i - \bar{y})^2}}$, simple linear regression has $R^2 = r^2$.

Proof [Supplement]. $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$, and $\text{TSS} = \text{RSS} + \sum(\hat{y}_i - \bar{y})^2$ (normal equations). So $R^2 = \hat{\beta}_1^2 S_{xx}/S_{yy} = S_{xy}^2/(S_{xx}S_{yy}) = r^2$. ■

Sales on TV: RSE $= 3.26$ (about 3,260 units), $R^2 = 0.612$, $F = 312.1$.

2 Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon, \qquad \texttt{sales} = \beta_0 + \beta_1 \texttt{TV} + \beta_2 \texttt{radio} + \beta_3 \texttt{newspaper} + \varepsilon.$$

β_j: average effect of a one-unit increase in X_j, *holding all other predictors fixed.*

2.1 Interpreting coefficients

Correlated predictors make interpretation hazardous and inflate all coefficient variances; uncorrelated (balanced) designs avoid this. Avoid causal claims for observational data.

Example: the woes of interpreting coefficients (Mosteller and Tukey, 1977)

Y = money in your pocket, X_1 = number of coins, X_2 = number of pennies, nickels and dimes. Alone, X_2 is positively associated with Y; given X_1, more small coins means fewer quarters, so its coefficient is negative.

2.2 Estimation

$\hat{\beta}$ minimizes $\text{RSS} = \sum_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip})^2$: $\hat{\beta} = (X^\top X)^{-1} X^\top y$ (normal equations $X^\top(y - X\hat{\beta}) = 0$). For $p = 2$ the fit is a plane.

	Coefficient	Std. error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

An extra \$1,000 on radio: about 189 more units at fixed TV and newspaper. Newspaper is significant alone but not here: it is correlated with radio and acts as a surrogate.

3 Four Important Questions

3.1 Is at least one predictor useful?

$H_0 : \beta_1 = \cdots = \beta_p = 0$ versus H_A: some $\beta_j \neq 0$,

$$F = \frac{(\mathrm{TSS} - \mathrm{RSS})/p}{\mathrm{RSS}/(n-p-1)} \;\sim\; F_{p,\,n-p-1} \text{ under } H_0.$$

Derivation: why F is the right statistic

$\mathbb{E}[\mathrm{RSS}/(n-p-1)] = \sigma^2$, and under H_0 also $\mathbb{E}[(\mathrm{TSS}-\mathrm{RSS})/p] = \sigma^2$ (Exercise 4.1); otherwise the numerator is larger. With Gaussian errors, $\mathrm{RSS}/\sigma^2 \sim \chi^2_{n-p-1}$ and $(\mathrm{TSS}-\mathrm{RSS})/\sigma^2 \sim \chi^2_p$ independently under H_0 (Cochran). With $p = 100$ and H_0 true, about 5 individual t-tests have $p < 0.05$ by chance; the F-test keeps its level.

Advertising: RSE $= 1.69$, $R^2 = 0.897$, $F = 570$.

3.2 Deciding on the important variables

Best subsets: 2^p models (over a billion for $p = 40$). **Forward selection** adds, one at a time, the most improving variable; **backward elimination** removes the least useful one. Each searches $1 + p(p+1)/2$ models (Ch. 6).

3.3 How well does the model fit?

$\mathrm{RSE} = \sqrt{\mathrm{RSS}/(n-p-1)}$, $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS}$, residual plots. R^2 never decreases with added predictors; RSE can increase.

3.4 How accurate is the prediction?

(i) **reducible error**: $\hat{Y} = \hat{\beta}_0 + \cdots + \hat{\beta}_p X_p$ estimates $f(X) = \beta_0 + \cdots + \beta_p X_p$ (confidence interval);

(ii) **model bias**: the linear model is an approximation;

(iii) **irreducible error** ε (prediction interval, always wider). For a new x_0: $\mathrm{Var}(x_0^\top \hat{\beta}) = \sigma^2 x_0^\top (X^\top X)^{-1} x_0$ and $\mathrm{Var}(Y_0 - x_0^\top \hat{\beta}) = \sigma^2\{1 + x_0^\top (X^\top X)^{-1} x_0\}$.

4 Qualitative Predictors

Categorical predictors (factors), e.g. student or marital status in the Credit data.

4.1 Two levels

$x_i = 1$ if person i owns a house, 0 otherwise:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{owner,} \\ \beta_0 + \varepsilon_i & \text{non-owner.} \end{cases}$$

Intercept 509.80 (SE 33.13), own[Yes] 19.73 (SE 46.05, $p = 0.669$): average debt \$509.80 versus \$529.53, not significantly different.

4.2 More than two levels

Region (East, South, West): $x_{i1} = \mathbb{I}(\text{South})$, $x_{i2} = \mathbb{I}(\text{West})$,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{South}, \\ \beta_0 + \beta_2 + \varepsilon_i & \text{West}, \\ \beta_0 + \varepsilon_i & \text{East (baseline)}. \end{cases}$$

One fewer dummy than levels (K dummies are collinear with the intercept). East \$531.00; South -18.69 ($p = 0.774$); West -12.50 ($p = 0.826$). Fitted values and the F-test do not depend on the baseline (Exercise 4.2).

5 The Gauss–Markov Theorem

Motivation. Under uncorrelated equal-variance errors, least squares has the smallest variance among linear unbiased estimators, without normality.

Theorem: Gauss–Markov

Let $y = X\beta + \varepsilon$ with $\mathbb{E}\varepsilon = 0$, $\mathrm{Var}\varepsilon = \sigma^2 I$, and X of full column rank. For any fixed c, among all linear unbiased estimators of $\theta = c^\top \beta$, the estimator $\hat{\theta} = c^\top \hat{\beta}$, $\hat{\beta} = (X^\top X)^{-1} X^\top y$, has the minimum variance, and it is unique. Least squares is the **BLUE**.

Proof [Supplement]. *Step 1 (unbiasedness constraint).* For $\tilde{\theta} = a^\top y$, $\mathbb{E}(a^\top y) = a^\top X\beta = c^\top \beta$ for all β iff $X^\top a = c$.

Step 2 (variance comparison). Least squares is $a_0 = X(X^\top X)^{-1} c$, with $X^\top a_0 = c$. Any unbiased $a = a_0 + d$ has $X^\top d = 0$, so $a_0^\top d = c^\top (X^\top X)^{-1} X^\top d = 0$ and

$$\mathrm{Var}(\tilde{\theta}) = \sigma^2 a^\top a = \sigma^2 (a_0^\top a_0 + d^\top d) \;\geq\; \sigma^2 a_0^\top a_0 = \mathrm{Var}(\hat{\theta}),$$

with equality iff $d = 0$. ■

Biased estimators such as ridge (Ch. 6) can have smaller mean squared error.

6 Extensions of the Linear Model

The standard model assumes **additive** and **linear** effects.

6.1 Interactions

If radio makes TV advertising more effective (*synergy*, or *interaction*), the TV effect is not a constant β_1. Advertising residuals: sales are overestimated when either budget is low and underestimated when the budget is split.

$$\texttt{sales} = \beta_0 + \beta_1 \texttt{TV} + \beta_2 \texttt{radio} + \beta_3 (\texttt{radio} \times \texttt{TV}) + \varepsilon.$$

	Coefficient	Std. error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

R^2: 89.7% → 96.8%, i.e. $(96.8 - 89.7)/(100 - 89.7) = 69\%$ of the remaining variability. An extra \$1,000 of TV: $(\hat{\beta}_1 + \hat{\beta}_3 \texttt{radio}) \times 1000 = 19 + 1.1\,\texttt{radio}$ units; of radio: $29 + 1.1\,\texttt{TV}$ units.

Hierarchy principle. With an interaction, include the main effects even if not significant; omitting them changes the meaning of the interaction (and makes it depend on the origin of the predictors).

6.2 Qualitative–quantitative interactions

$$\texttt{balance}_i \approx \beta_0 + \beta_1 \texttt{income}_i + \begin{cases} \beta_2 + \beta_3 \texttt{income}_i & \text{student} \\ 0 & \text{non-student} \end{cases}$$

$$= \begin{cases} (\beta_0 + \beta_2) + (\beta_1 + \beta_3)\texttt{income}_i & \text{student,} \\ \beta_0 + \beta_1 \texttt{income}_i & \text{non-student :} \end{cases}$$

different intercepts *and* slopes.

6.3 Nonlinear effects: polynomial regression

Auto data: $\texttt{mpg} = \beta_0 + \beta_1 \texttt{hp} + \beta_2 \texttt{hp}^2 + \varepsilon$, estimates 56.9001, −0.4662, 0.0012 (all $p < 0.0001$). Still *linear* in the coefficients. Residuals versus fitted: strong U-shape for the linear fit, little pattern for the quadratic.

Summary

- OLS minimizes RSS; SEs, t-tests and CIs quantify coefficient uncertainty; RSE and R^2 measure fit.
- F-test: is any predictor useful; dummies encode factors relative to a baseline.
- Gauss–Markov: OLS is BLUE. Interactions and polynomials stay within least squares.

7 Selected exercises

Motivation. The expectations behind F, baseline invariance of dummy coding, and ISLR Ex. 3.3.

Exercise 4.1 (Expected residual and regression sums of squares)

Let $y = X\beta + \varepsilon$ with $\mathbb{E}\varepsilon = 0$, $\mathrm{Var}\varepsilon = \sigma^2 I_n$, and $X \in \mathbb{R}^{n\times(p+1)}$ of full column rank with first column $\mathbf{1}$.
(a) Show that $\mathbb{E}[\mathrm{RSS}] = (n - p - 1)\sigma^2$, so RSE^2 is unbiased for σ^2.
(b) Show that under $H_0 : \beta_1 = \cdots = \beta_p = 0$, $\mathbb{E}[\mathrm{TSS} - \mathrm{RSS}] = p\sigma^2$.

Solution [Supplement].

(a)

Step 1. $H = X(X^\top X)^{-1}X^\top$ is a symmetric idempotent with $HX = X$, so $(I-H)y = (I-H)\varepsilon$ and $\mathrm{RSS} = \varepsilon^\top(I-H)\varepsilon$.

Step 2. For fixed M, $\mathbb{E}[\varepsilon^\top M\varepsilon] = \mathbb{E}\mathrm{tr}(M\varepsilon\varepsilon^\top) = \sigma^2\mathrm{tr}M$.

Step 3. $\mathrm{tr}(I-H) = n - \mathrm{tr}\big((X^\top X)^{-1}X^\top X\big) = n-(p+1)$.

(b) $J = \mathbf{1}\mathbf{1}^\top/n$ projects onto $\mathrm{span}(\mathbf{1}) \subseteq \mathrm{col}(X)$, so $HJ = JH = J$ and $H-J$ is a projection of trace p. $\mathrm{TSS}-\mathrm{RSS} = y^\top(I-J)y - y^\top(I-H)y = y^\top(H-J)y$. Under H_0, $y = \beta_0\mathbf{1}+\varepsilon$ and $(H-J)\mathbf{1} = 0$, so this is $\varepsilon^\top(H-J)\varepsilon$, with expectation $p\sigma^2$ by Step 2. ■

Exercise 4.2 (The baseline does not matter)

A factor has levels $1,\dots,K$, each observed at least once, with indicator columns $D_1,\dots,D_K \in \mathbb{R}^n$. Design A uses $\mathbf{1}, D_2,\dots,D_K$ (baseline 1); design B uses $\mathbf{1}$ and all D_k with $k \neq r$ (baseline r); other predictors, if any, are the same in both.
(a) Show that A and B have the same column space, hence the same fitted values, RSS and overall F-statistic.
(b) Express the coefficients of B through those of A.

Solution [Supplement].

(a) $\sum_{k=1}^K D_k = \mathbf{1}$, so $D_1 = \mathbf{1} - \sum_{k\geq 2} D_k \in \mathrm{col}(A)$ and $\mathrm{col}(A) = \mathrm{span}(D_1,\dots,D_K,\text{others})$; likewise for $\mathrm{col}(B)$. Equal column spaces give the same projection, $\hat{y}$ and RSS; TSS, n and the number of predictors agree, so F agrees.

(b) With no other predictors, set $\beta_1^A := 0$, $\beta_r^B := 0$. By (a), $\beta_0^A + \beta_k^A = \beta_0^B + \beta_k^B$ for each level k (coefficients are unique by full column rank). $k = r$ gives $\beta_0^B = \beta_0^A + \beta_r^A$, hence $\beta_k^B = \beta_k^A - \beta_r^A$. Credit example with baseline South: intercept 512.31, East +18.69, West +6.19. ■

Exercise 4.3 (Reading an interaction model; ISLR Ex. 3.3)

Starting salary (thousands of dollars) is regressed on $X_1 = \mathrm{GPA}$, $X_2 = \mathrm{IQ}$, $X_3 = \mathrm{Level}$ (1 college, 0 high school), $X_4 = X_1X_2$ and $X_5 = X_1X_3$, giving $\hat{\beta}_0 = 50$, $\hat{\beta}_1 = 20$, $\hat{\beta}_2 = 0.07$, $\hat{\beta}_3 = 35$, $\hat{\beta}_4 = 0.01$, $\hat{\beta}_5 = -10$.
(a) For fixed IQ and GPA, who earns more on average, college or high-school graduates?
(b) Predict the salary of a college graduate with IQ 110 and GPA 4.0.
(c) True or false: since $\hat{\beta}_4$ is very small, there is little evidence of a GPA–IQ interaction.

Solution.

(a) College minus high school: $\hat{\beta}_3 + \hat{\beta}_5\,\mathrm{GPA} = 35 - 10\,\mathrm{GPA}$, positive iff $\mathrm{GPA} < 3.5$. High-school graduates earn more if $\mathrm{GPA} > 3.5$; below that, college graduates.

(b) $50 + 20(4) + 0.07(110) + 35 + 0.01(4)(110) - 10(4) = 50 + 80 + 7.7 + 35 + 4.4 - 40 = 137.1$, about \$137,100.

(c) False. $\mathrm{GPA}\times\mathrm{IQ} \approx 400$, so 0.01 per unit is worth about 4 thousand dollars; evidence is measured by $\hat{\beta}_4/\mathrm{SE}(\hat{\beta}_4)$, which is not given. ■